INTRODUCTION

In recent years, the changes in the way we live our lives and the environment we operate in have immensely shifted. From a simplistic lifestyle to a more scientific and technological driven format. Everything around us is becoming increasingly more artificial than natural. Trying to be current and advance with science and technology, operating in urbanized environments can cause a lot of stress and mental health instability. However, biophilic interventions serve as an avenue for people to reset and stabilize, providing natural elements into built environments to improve mental health and reduce stress.

Recent literature has increasingly highlighted the therapeutic potential of biophilic design in enhancing mental wellbeing, particularly in urban and clinical environments where access to nature is limited. Studies such as those by Gillis & Gatersleben (2015) and Kellert (2008) emphasize the psychological benefits of integrating natural elements into built environments, including reduced stress, improved mood, and enhanced cognitive function. Research on virtual reality (VR) therapy such as Bell et al. (2024) and Park et al. (2019) demonstrates how immersive digital environments can simulate nature effectively, offering scalable and accessible interventions for mental health.

In a broader view, due to the recent increase in mental health importance, remedies to improve mental health has become a high priority. The digital lifestyle we live in today has limited access to greenery as well as compact urbanized environments. Biophilic interventions acts as a means to reconnect people with natural environments, thus reducing stress and improving mental health.

Taking into consideration of environmental psychology, urban planning and mental healthcare, this research project strives to understand psychology behind the connection between humans and nature as well as understanding the how that connection improves the mental health. Based on the findings the project can be used as a foundation to

advocate for biophilic designs to be considered as high priority when it comes to mental health.

AIM AND OBJECTIVES

Aim

This project aims to evaluate the human responses towards virtual reality being used to build biophilic environments that promote mental wellness and reduce stress. this project also seeks to see, based on results, the effectiveness of these virtual environments and their potential of them being used for psychological rehabilitation.

Objectives

Following the SMART (Specific, Measurable, Achievable, Realistic and Time Bound) methodology the objectives of the project are;

1. Specific: Develop a various VR Environments (e.g., Forest, Ocean, Garden) that imitates natural environments that can be used for psychological rehabilitation. For this project, research would be made the discover specific inclusions in the environments listed above that stimulate the mind allowing people to be relaxed and comfortable which in return promotes mental wellbeing and why it works. With the information gotten as a guide, the VR environments would be tailored to meet the requirements of having those content in the VR environment.
2. Measurable: Evaluate the impact of the VR environments on participants mental wellbeing and heart rate using validated psychological tools such as PSS (Perceived Stress Scale), PANAS (Positive and Negative Affected Schedule) and a heart rate monitor as a biometric indicator. Obtaining these information helps determine the effectiveness of not only biophilic interventions, but the effectiveness of VR as a means to build biophilic environments.

3. Achievable: Conduct a well monitored study using a minimum of 15 participants of different ages and backgrounds for potential diversity in results. Having numerous participants with different age groups and backgrounds for a variety of results and provides more global statistics, allowing the project to be relevant on a more global scale.
4. Relevant: the project would be in accordance with the current mental health needs in the society. The increase in mental health problems is alarming. So much so that remedies are being sorted out for. This project would aid the search for mental health rehabilitation tools. Tools that are cost effective compared to the practical biophilic environments.
5. Time-bound: Complete the entirety of the project within a 7-month timeframe including prototypes and the results from the participants. Time management is very essential in the medical or medical related fields. The more effective and efficient the project is the more it can provide relevance to the society.

TASKS AND DELIVERABLES

The table below provides details on the milestones, tasks, deliverables and timeline dedicated to this project;

Milestone	Tasks	deliverables	Timeline
Project Initiation	- literature review - Tool selection	- Review similar and or related findings on project topic	Month 1

	-scope definition	-Obtain tools and licensing from the university. - Project proposal, initial Gantt chart	
Review Point 2	-Finalize design specifications -Begin prototype development	-Acquire all the needed tools and software. -Start designing environment based on the information gotten during research.	Month 2
Prototype Development	-Build VR environments -Collect initial data	-Using the finalized designs, craft a VR environment. -Document the data gotten from pilot test and confirm the environments meet the requirements.	Month 3-4
Participant Recruitment	-Gather participants	-Consensually recruit individuals through social media and flyers.	Month 4
Testing Phase	-conduct controlled study, document	-Tabulate obtained data from the	Month 5

	psychological and biometric data	participants and log all observations after testing.	
Data Analysis	-Analize results	-Using statistical methods, analyse the obtained data and design a chart and graph showing the information.	Month 5-6
Demonstration	-Present findings	-with a live demo and documentation, the project would be demonstrated to the university.	Month 6
Project Thesis	-Final thesis documentation	-Compile full report including thesis and submit to the university.	Month 7

Out of Scope Deliverables

These are explicitly excluded from the project:

- Acquisition of hardware (VR headsets).
- Longitudinal studies beyond the 7-month timeframe.
- Clinical trials or medical certification.
- Integration with commercial platforms or apps.

Functionality

- Immersive VR environments with nature-based visuals and ambient sounds.

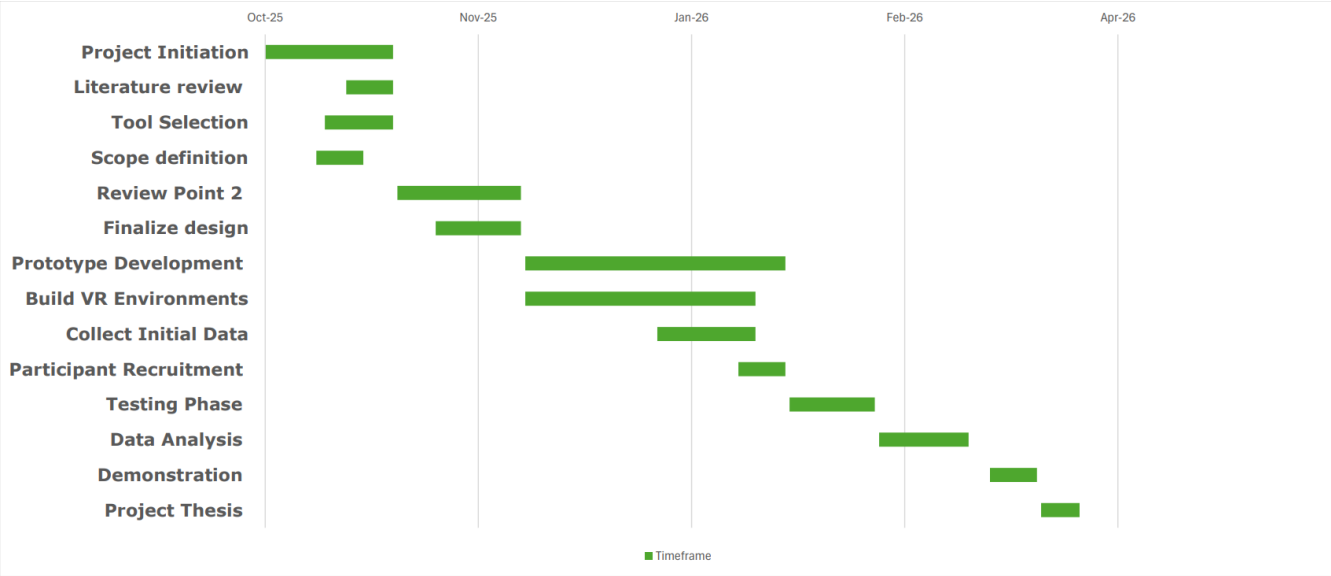
- User interaction tracking (e.g., gaze, movement).
- Data collection modules for psychological and biometric metrics.

Out of Scope Functionality

These functionalities are in the case that the project is further pursued;

- Multiplayer VR features.
- AI-driven personalization of environments (within guidelines).
- Real-time biofeedback adaptation.
- Mobile versions of the VR experience.

GANTT CHART



RESOURCES

Resources to Meet the Scope of the Project;

1. Technological Resources

- VR Development Software: Unreal Engine for building biophilic environments.

    - Justification: These platforms support high-fidelity graphics and integration with biometric sensors.
- Biometric Sensors: Heart rate monitor (Polar H10).
    - Justification: Needed for physiological data collection to assess stress levels. Polar H10 is recognised as the most accurate heart rate monitor.
- Psychological Assessment Tools: PSS (Perceived Stress Scale), PANAS (Positive and Negative Affect Schedule).
    - Justification: Validated tools for measuring mental wellbeing.
- Data Analysis Software: Kaggle (with libraries like pandas, matplotlib, seaborn).
    - Justification: Required for statistical analysis of collected data.

2. Human Resources

- Participants: Minimum of 15 individuals from diverse backgrounds.
    - Justification: Ensures generalizability and inclusivity.
- Research Supervisors / Advisors: Experts in psychology, VR development, and healthcare.
    - Justification: For guidance, ethical oversight, and validation.
- Ethics Committee: Institutional review board for approval.
    - Justification: Ensures ethical compliance in human studies.

3. Documentation & Reporting Tools

- Gantt Chart Software: Microsoft Excel.
    - Justification: For tracking milestones and deliverables.
- Thesis Writing Tools: Microsoft Word.
    - Justification: For compiling and formatting the final report.

Sources of Information Required to Conduct the Project

1. Academic Literature

reviewed journals on:

- Biophilic design (Journal of Environmental Psychology)
- Virtual reality in therapy (Cyberpsychology, Behavior, and Social Networking)
- Mental health interventions (The Lancet Psychiatry)
    - Justification: Provides theoretical foundation and evidence-based practices.

2. Books

- "Biophilic Design: The Theory, Science and Practice of Bringing Buildings to Life" by Stephen Kellert.
- "Environmental Psychology: Principles and Practice" by Robert Gifford.
    - Justification: Offers deep insights into the psychological and design principles behind biophilic interventions.

3. Government and NGO Reports

- WHO reports on mental health and urban wellbeing.
- NHS or local health authority guidelines on psychological rehabilitation.
    - Justification: Aligns project relevance with public health priorities.

4. Technical Documentation

- VR SDK documentation (Oculus SDK).
- Sensor integration manuals.
    - Justification: Ensures smooth technical implementation.

5. Case Studies and White Papers

- Existing VR therapy application (Limbix).
- Urban biophilic design implementations.
    - Justification: Helps benchmark and refine your prototype.

PROJECT RISKS

Risk Area	Potential Risk	Justification	Recovery Strategy
Technical Development	VR prototype may malfunction or fail to meet design expectations.	Complex integration of visuals, interaction tracking, and biometric tools may exceed skill/time limits.	Use agile development, iterative testing, and community support.
Participant Recruitment	Difficulty recruiting 15 diverse participants.	Limited outreach, ethical delays, or lack of interest may reduce required participants.	Start recruitment early, use multiple channels, offer incentives, and prepare backup list.
Ethical Approval	Delays or rejection from ethics committee.	Human studies with biometric and psychological data require strict review.	Submit early, consult ethics advisor, prepare alternative study designs in case of rejection.
Data Collection	Incomplete or inconsistent participant data.	Participants may not follow instructions or sensors may malfunction.	Conduct pilot testing, train participants, and use redundant data collection methods.
Time Constraints	Project may exceed the 7-month timeline.	Multiple interdependent phases may take longer than expected.	Use detailed Gantt chart, prioritize tasks, and limit scope if needed.
Software Compatibility	VR software may not integrate with biometric tools.	SDK/API conflicts or advanced coding requirements may arise.	Choose widely supported platforms, consult experts, and use modular design.
Data Privacy & Security	Risk of breach or misuse of sensitive data.	Biometric and psychological data are sensitive and must be protected.	Encrypt data, anonymize records, and follow GDPR/university policies.
User Experience	VR environments may cause discomfort (motion sickness).	Poor design or hardware.	conduct usability testing and allow exit options.
Hardware Access	Limited access to VR headsets or biometric devices.	Hardware acquisition is out of scope and may restrict testing.	Borrow a headset from university.

CAREER ASPIRATIONS

This project aligns closely with my career aspirations in immersive technology development. By focusing on virtual reality (VR) as a medium for biophilic interventions, I am positioning myself within a rapidly growing field that combines psychology, technology, and design to address pressing global health challenges in this digital world as well as giving me advanced understanding of immersive technology development.

Possible careers

1. Digital Mental Health Specialist

- Role: Develop and implement technology-based mental health solutions.

- Skills from Project: VR therapy design, psychological assessment, biometric data analysis.

2. UX/UI Designer for Health Applications

- Role: Design user-friendly interfaces for therapeutic and wellness platforms.
- Skills from Project: User interaction tracking, immersive environment design, usability testing

3. VR Developer for Therapeutic Applications

- Role: Build immersive experiences for rehabilitation, therapy, or education.
- Skills from Project: Unreal Engine development, biometric sensor integration, prototype testing.

4. Public Health Technologist or Policy Advisor

- Role: Advocate for and implement digital mental health strategies in public health.
- Skills from Project: Alignment with mental health policies, social impact analysis.

LSEPIs Related to Project Outcomes

1. Legal Issues
    - Data Protection Act & UK GDPR: The project collects sensitive biometric and psychological data. Compliance requires:
    - Informed consent.
    - Secure data storage and encryption.
    - Anonymization of participant data.
    - Intellectual Property: All third-party assets used in the VR prototype (e.g., textures, soundscapes) must be properly licensed.
    - Computer Misuse Act: Ensures that software and data handling do not violate system integrity or misuse computing resources.
2. Social Issues

- The project addresses mental health accessibility, especially for:
- Urban populations with limited access to nature.
- Clinics lacking therapeutic resources.
- People facing barriers to mental health support.

3. Ethical Issues
   - Health and Safety: VR environments must be designed to avoid motion sickness or psychological discomfort.
   - Safeguarding: While DBS checks may not be required for adult participants, ethical oversight ensures participant safety.
   - Risk Assessment: Includes usability testing, exit options, and pilot studies to minimize harm.
   - Transparency and Consent: Participants must be fully informed of the study's purpose, data usage, and their rights.
4. Professional Issues
   - The project adheres to the BCS Code of Conduct, ensuring:
   - Integrity in data handling and reporting.
   - Respect for participant dignity and autonomy.
   - Professional competence in VR development and psychological research.
   - Collaboration with supervisors ensures professionalism.

LSEPIs Related to the Research Process

1. Legal Compliance During Research
   - Ethical approval is required from the university's ethics committee before participant recruitment.
   - All data collection tools (e.g., PSS, PANAS, heart rate monitors) must be used in accordance with legal and institutional guidelines.
2. Social Impact of Research
   - The recruitment strategy emphasizes diversity and inclusion.

- The research may influence future mental health interventions, contributing to public health policy and digital therapy practices.

3. Ethical Conduct in Research
    - Participants must be treated with respect and care throughout the study.
    - Data must be anonymized and securely stored to protect privacy.
4. Professional Standards in Research
    - The project follows academic research protocols for study design, data analysis, and reporting.
    - All communications and documentation reflect professionalism and academic integrity.

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